

BROADVIEW, SK, Canada

WMO: 718610

Lat: **50.368N**Lon: **102.571W**Elev: **600**StdP: **94.32**Time Zone: **-6.00 (W06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.7	-29.9	-37.1	0.1	-32.6	-34.2	0.2	-29.8	13.2	-7.7	11.8	-8.4	3.4	300	0.705

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	29.2	19.2	27.4	18.4	25.8	17.6	21.2	26.6	19.8	25.3	18.6	24.2	5.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	15.1	24.4	17.8	13.7	22.8	16.6	12.7	21.4	64.5	26.7	59.4	25.2	55.5	24.3	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.3	9.1	-36.9	33.1	3.2	1.9	-39.1	34.4	-41.0	35.6	-42.7	36.6	-45.0	38.0		
(5)			-37.0	23.5	3.1	1.5	-39.2	24.5	-41.0	25.4	-42.8	26.2	-45.0	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-14.8	-13.2	-6.4	3.2	10.2	15.5	18.3	17.0	12.1	4.2	-4.8	-12.2	(6)
(7)		DBStd	13.32	8.72	8.09	7.95	6.07	4.68	3.54	3.05	3.43	4.51	5.13	6.89	8.14	(7)
(8)		HDD10.0	3551	768	649	509	214	56	3	0	1	28	190	445	688	(8)
(9)		HDD18.3	5863	1026	883	768	455	255	98	39	68	191	439	695	947	(9)
(10)		CDD10.0	811	0	0	0	9	61	167	256	216	92	10	0	0	(10)
(11)		CDD18.3	81	0	0	0	0	2	12	37	25	4	0	0	0	(11)
(12)		CDH23.3	1028	0	0	0	3	53	144	388	326	110	3	0	0	(12)
(13)		CDH26.7	234	0	0	0	0	8	27	88	89	22	1	0	0	(13)
(14)	Wind	WSAvg	4.2	4.4	4.3	4.6	4.6	4.0	3.6	3.5	4.0	4.4	4.4	4.3	(14)	
(15)	Precipitation	PrecAvg	416	18	13	22	27	48	62	56	56	27	16	19	(15)	
(16)		PrecMax	550	37	32	58	108	121	131	114	155	145	67	35	33	(16)
(17)		PrecMin	294	3	4	4	0	7	8	12	8	3	0	2	7	(17)
(18)		PrecStd	61	9	7	13	23	30	34	28	38	37	22	9	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	4.3	13.6	22.9	27.8	29.4	31.2	31.9	29.3	22.2	14.6	5.7	(19)
(20)			MCWB	1.5	2.0	7.6	11.9	15.8	20.0	21.3	19.4	17.3	12.6	8.0	2.2	(20)
(21)		2%	DB	2.1	2.1	9.1	18.7	24.6	26.9	29.1	29.1	26.5	18.3	9.4	3.3	(21)
(22)			MCWB	0.3	0.3	4.7	9.8	13.6	17.6	20.3	18.8	16.2	10.8	4.8	0.9	(22)
(23)		5%	DB	0.0	0.3	5.7	16.0	21.9	24.8	27.2	26.9	23.8	14.9	6.4	1.1	(23)
(24)			MCWB	-1.3	-0.9	3.0	8.1	12.3	16.6	19.5	18.4	15.1	9.0	3.0	-0.4	(24)
(25)		10%	DB	-2.3	-1.6	3.2	13.1	19.3	22.9	25.5	24.7	20.9	12.2	3.9	-1.2	(25)
(26)			MCWB	-3.3	-2.7	1.3	6.8	11.1	15.6	18.6	17.2	13.6	7.8	1.7	-2.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	2.2	7.8	12.5	16.8	22.3	23.3	22.1	19.3	13.9	8.2	2.6	(27)
(28)			MCD B	3.7	4.0	12.6	20.9	22.9	26.8	28.4	28.0	26.0	20.4	14.1	5.3	(28)
(29)		2%	WB	0.4	0.5	5.1	10.6	14.7	19.2	21.7	20.4	17.4	11.5	5.2	1.0	(29)
(30)			MCD B	2.1	1.8	8.7	17.7	21.8	24.5	26.9	26.6	24.0	16.7	8.7	3.0	(30)
(31)		5%	WB	-1.2	-0.9	3.0	8.7	13.3	17.8	20.4	19.1	15.8	9.7	3.3	-0.5	(31)
(32)			MCD B	0.0	0.3	5.4	14.7	20.1	22.9	25.6	25.2	22.3	14.0	5.9	1.1	(32)
(33)		10%	WB	-3.2	-2.7	1.4	7.2	12.0	16.4	19.2	17.9	14.2	8.1	1.8	-2.3	(33)
(34)			MCD B	-2.3	-1.6	3.1	12.6	17.8	21.4	24.5	23.4	19.8	11.9	3.8	-1.1	(34)
(35)	Mean Daily Temperature Range	MDBR	9.6	9.8	9.8	11.4	12.5	10.9	11.8	12.9	12.8	10.5	9.0	9.0	(35)	
(36)		5% DB	MCD BR	10.3	9.1	11.6	16.4	16.8	14.2	14.4	16.3	17.5	15.8	12.6	10.5	(36)
(37)			MCWBR	9.3	7.8	8.2	9.0	8.0	6.9	7.5	8.0	8.9	9.3	8.6	8.8	(37)
(38)		5% WB	MCD BR	10.2	8.9	11.3	15.3	14.3	12.0	12.9	14.4	15.6	14.3	11.7	10.4	(38)
(39)			MCWBR	9.2	7.7	8.2	8.8	7.4	6.7	7.5	7.9	8.5	8.7	8.3	8.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.261	0.249	0.274	0.332	0.340	0.358	0.367	0.382	0.341	0.282	0.247	0.225	(40)	
(41)		taud	2.245	2.326	2.432	2.379	2.412	2.393	2.389	2.330	2.424	2.563	2.502	2.388	(41)	
(42)		Ebn at Noon	776	908	944	912	916	899	883	849	853	865	821	797	(42)	
(43)		Edh at Noon	69	83	91	109	110	114	112	114	93	66	54	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.30	2.34	3.68	4.89	5.82	5.99	6.50	5.16	3.89	2.28	1.26	0.93	(44)	
(45)		RadStd	0.11	0.22	0.28	0.32	0.48	0.44	0.36	0.41	0.33	0.28	0.18	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page